Check for updates

ChemistrySelect

Research Article
doi.org/10.1002/slct.202300703

Chemistry Europe
European Chemical Societies Publishing

www.chemistryselect.org

Facile Synthesis of Glycidyl Azide Polymer (GAP) in a Tetrabutylammonium Bromide/Ethylene Glycol Based Deep Eutectic Solvent

Jung-Ah Shin,*[a] Jihye Jung,[b] Youngtaek Moon,[b] and Sejin Lee[b]

Glycidyl azide polymer (GAP) is an energetic material used in composite solid rocket propellants. Herein, a novel synthetic method for GAP was introduced using a tetrabutylammonium bromide and ethylene glycol-based deep eutectic solvent (DES). Using this DES combination, the azidation of poly(epichlorohydrin) was successfully conducted within several hours. This strategy eliminates the use of organic solvents during the entire synthetic process and provides the advantage of recovery and reuse of the DES.

Introduction

Glycidyl azide polymer (GAP), one of the most widely known energetic binders, is an energetic polymer with high density and high heat generation (314–398 kJ/azido group).[1] It is an energetic, insensitive, and thermally stable material, particularly useful in composite solid rocket propellants.[2] GAP can be readily synthesized by azidation of poly(epichlorohydrin) (PECH). Under typical reaction conditions, GAP is obtained by nucleophilic substitution of PECH with sodium azide in dimethylsulfoxide (DMSO) or *N*,*N*-dimethylformamide (DMF) at 100 °C.[2b,3] Moreover, during workup, this process requires the use of organic solvents such as dichloromethane or ethyl acetate to purify GAP. However, this results in environmental pollution and harm to human health. In addition, it is tedious work, taking a long time for the organic layer and the aqueous layer to separate during workup, and it is also difficult to remove DMF (or DMSO) from the GAP cleanly. Therefore, to overcome the drawbacks of organic solvents, researchers have explored effective and eco-friendly alternative strategies to obtain GAP from PECH.

Wagner's group reported azidation with molten quaternary ammonium azide salts, such as tetrabutylammonium azide (TBAA).[4] In 2016, Xu et al. developed a synthetic method for GAP in mixed solvents of different mass ratios with the ionic liquid 1-butyl-3-methylimidazolium chloride ([Bmim]Cl) and water.[5] Kasztankiewicz and Frankel also reported the synthesis of GAP by treating PECH with NaN_3 in water above 90 °C in the presence of catalytic amounts of quaternary ammonium salt as a phase transfer catalyst (PTC). However, the reaction was not complete after 24 h.[1a,6]

Recently, owing to their potential applications in various fields, deep eutectic solvents (DESs) have gained considerable attention as an emerging class of green liquids. DESs are homogeneous mixtures formed by the combination of a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA), affording a lower melting point than that of both individual components.[7] They have properties similar to ionic liquids (ILs), but are relatively cheaper, less toxic, recyclable, and biodegradable.[8] Because of the possibility of improving synthetic processes owing to their reusability and eco-friendliness, DESs have been employed as novel reaction media to replace typical organic solvents in various polymer syntheses.[9]

In our pioneering studies on developing green synthetic methods for energetic materials in DESs, we recently reported a $ZnCl_2$-catalyzed synthesis of 5-substituted-1H-tetrazoles from organic nitriles and NaN_3 in a Betaine-based DES, which plays the dual role of a catalyst and an environmentally benign reaction medium.[10]

As part of our continuing work on the DES-based development of readily applicable synthetic processes, herein, we report the synthesis of GAP using PECH with NaN_3 in a recyclable and environmentally benign tetrabutylammonium bromide(TBAB)-ethylene glycol(EG)-based DES.

Results and Discussion

First, as shown in Figure 1, representative HBAs were selected: choline chloride (ChCl), betaine (Bet), tetrabutylammonium halides (TBAB, TBAC, and TBAI), tetraethylammonium bromide (TEAB), dodecyltrimethylammonium chloride (DTAC), and methyltrioctylammonium chloride (MOAC). The HBDs selected were urea, ethylene glycol (EG), glycerol (Gly), 1,4-butanediol (BD), 1,8-octanediol (OD), and malic acid (MA). The DESs were synthesized by mixing the HBAs and HBDs in a molar ratio of 1:2. The mixtures were heated to 100 °C with stirring to

[a] Dr. J.-A. Shin
The Missile R&D Institute 2nd Directorate, Agency for Defense Development, Daejeon 34186, Korea
E-mail: zzanga0491@add.re.kr

[b] Dr. J. Jung, Dr. Y. Moon, Dr. S. Lee
The Missile R&D Institute 2nd Directorate, Agency for Defense Development, Daejeon 34186, Korea

Supporting information for this article is available on the WWW under https://doi.org/10.1002/slct.202300703

© 2023 Wiley-VCH GmbH

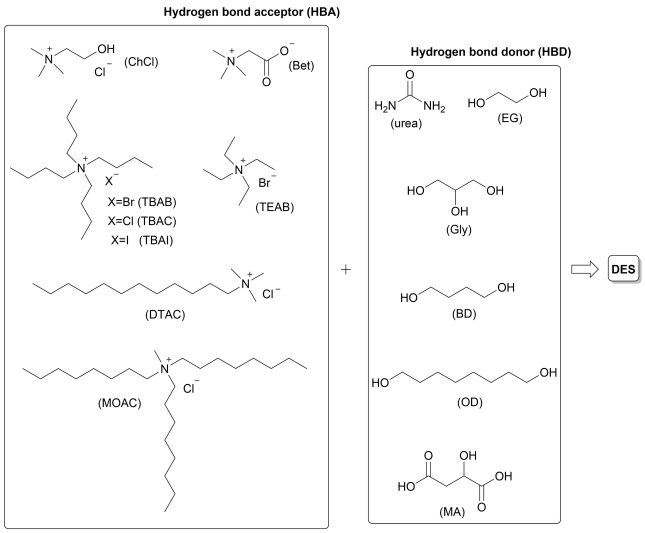


Figure 1. General scheme of DESs prepared by mixing HBAs with HBDs.

produce clear homogenized solutions, and the resulting molten salts were used directly in the reaction without purification.

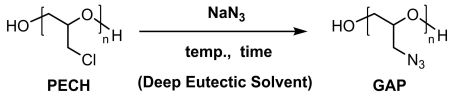
To examine the effects of the DESs on the azidation of PECH, reactions were carried out with PECH and NaN_3 in various eutectic solvents.

The solvents were prepared as described above, and the results of the reactions are presented in Table 1. The reaction hardly proceeded in the combination of ChCl/urea, a widely known DES system (Table 1, entry 1). Poor conversions were also obtained when the reaction was performed using ChCl/Gly and ChCl/EG (Table 1, entries 2 and 3).

Using Bet as the HBA component instead of ChCl was also ineffective in generating the GAP adduct (Table 1, entries 4 and 5). Interestingly, when TBAI was added in a catalytic amount to the ChCl/EG combination, the conversion was greatly improved to 81% (Table 1, entry 6). Thus, we also decided to investigate the combination of DES with a tetraalkylammonium source as the HBA element. Using the combination of TBAB and diols, such as EG, Gly, BD, and OD, PECH was successfully converted to GAP with observed conversions of 81–95% in all cases; in particular, the TBAB/EG system showed the best conversion of 95% (Table 1, entries 7–10). In the presence of TBAB and MA, the conversion was only 35% (Table 1, entry 11). Similarly, other tetrabutylammonium salts, such as TBAC and TBAI, were also highly efficient in the reaction, although they were less cost-effective than TBAB or proved difficult to remove during the workup process (Table 1, entries 12 and 13). To examine the effect of the alkyl group of the ammonium salts, TEAB, DTAC, and MOAC were tested; the conversion to GAP was unsatisfactory compared to TBAB/EG (Table 1, entries 14–16). Through these preliminary studies, we maximized the reaction efficiency by selecting suitable HBA-HBD combinations that can dissolve both PECH and NaN_3.

Interestingly, we observed that the viscous PECH did not dissolve uniformly in the ChCl-based DESs, whereas it easily dissolved in the TBAB-based DESs. Thus, we decided to

Table 1. Results for azidation of PECH in various conditions.[a]



Entry	DES	Additive	Time	Conversion[b]
1	ChCl/urea	–	24 h	<5%
2	ChCl/Gly	–	24 h	<5%
3	ChCl/EG	–	24 h	<5%
4	Bet/Gly	–	24 h	<5%
5	Bet/EG	–	24 h	<5%
6	ChCl/EG	TBAI 0.5 eq.	24 h	81%
7	TBAB/EG	–	5 h	95%
8	TBAB/Gly	–	5 h	81%
9	TBAB/BD	–	5 h	93%
10	TBAB/OD	–	5 h	90%
11	TBAB/MA	–	5 h	35%
12	TBAC/EG	–	5 h	95%
13	TBAI/EG	–	5 h	93%
14	TEAB/EG	–	5 h	57%
15	DTAC/EG	–	5 h	81%
16	MOAC/EG	–	5 h	23%

[a] All the reactions were performed using PECH 0.3 g (0.32 mmol), NaN_3 (1.2 equiv.) in DES (2 mL) at 100 °C for 18 h. [b] The conversion ratio of the crude compound was determined by FT-IR spectroscopy: ChCl (choline chloride), Bet (betaine), TBAB (tetrabutylammonium bromide), TBAC (tetrabutylammonium chloride), TBAI (tetrabutylammonium iodide), TEAB (tetraethylammonium bromide), DTAC (dodecyltrimethylammonium chloride), MOAC (methyltrioctylammonium chloride), Gly (glycerol), EG (ethylene glycol), Im (imidazole), OD (1,8-octanediol), MA (malic acid), BD (1,4-butanediol).

investigate these DES combinations in detail in terms of normalized polarity (E_T^N) and Kamlet-Taft parameters (α, β, and π^*) to compare their solvatochromic characteristics.[11] $E_T(33)$, as a measurement value of the solvent's overall polarity arising from interactions between the dye and solvent, is ultimately converted to E_T^N values that are between 0 for TMS (extremely non-polar) and 1 for water (extremely polar) (Table 2 and SI). ChCl-based DESs, which are hydrophilic, showed significantly higher E_T^N values than betaine- or TBAB-based DESs.

The E_T^N values of TBAB-diol combinations such as TBAB/EG, TBAB/BD, and TBAB/OD were lower than those of other DESs, indicating lower dipolarity/polarizability and/or HBD acidity of the TBAB-diol DESs. Combining the results of Tables 1 and 2, these E_T^N values proved that in most TBAB-based DESs with low E_T^N values, non-polar PECH was dissolved and effectively converted to GAP.

The conversions of the reactions were confirmed by FT-IR analysis in all cases. For PECH, a characteristic absorption peak was observed at 745 cm^{-1} based on the C–Cl stretching vibration. As the reaction progressed, this peak became weaker and finally disappeared, indicating the change of chloride groups into azide groups. A strong absorption at 2095 cm^{-1} was observed due to the asymmetric N_3 stretching vibration

© 2023 Wiley-VCH GmbH

23656549, 2023, 30, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/slct.202300703 by Zhejiang University, Wiley Online Library on [03/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. E_T^N and Kamlet–Taft parameters for all DESs.

Entry	DES	λ_{max}(33) (nm)[a]	E_T(33) (kcal/mol)	E_T^N[b]
1	ChCl/urea	438	65.28	0.81
2	ChCl/Gly	432	66.26	0.84
3	ChCl/EG	436	65.65	0.82
4	Bet/Gly	441	64.83	0.79
5	Bet/EG	464	61.62	0.69
6	TBAB/EG	472	60.64	0.66
7	TBAB/Gly	462	61.95	0.71
8	TBAB/BD	488	58.59	0.60
9	TBAB/OD	490	58.35	0.59
10	TBAC/EG	482	59.32	0.62
11	TBAI/EG	456	62.70	0.73
12	TEAB/EG	451	63.47	0.75
13	DTAC/EG	471	60.70	0.67
14	MOAC/EG	391	73.12	1.05

[a] UV-Vis absorbance maxima (λ_{max}) of betaine dye 33 for all DESs at 37 °C. [b] E_T(30) values and Kamlet-Taft parameters (α, β, and π*) SEE SI. E_T^N values were calculated using an E_T(30)-based equation [11].

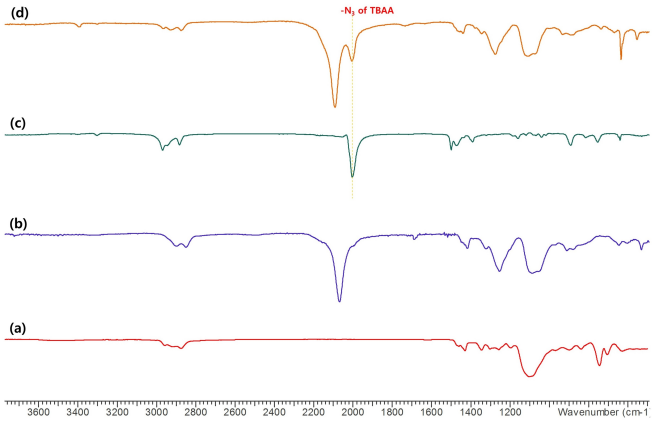
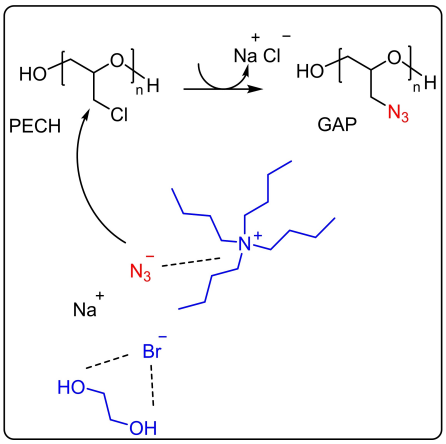


Figure 2. FT-IR spectra of the reaction process (a) PECH, (b) GAP, (c) TBAA, (d) trace of TBAA in the reaction mixture.



Scheme 1. Plausible mechanism for the synthesis of GAP in DES composed of TBAB/EG.

(Figure 2). Interestingly, as depicted in Figure 2(d), TBAA (tetrabutylammonium azide) derived from TBAB and NaN_3 was observed during the reaction. Under TBAB-EG system, Br^- moved further away from TBA^+ and formed a hydrogen bond with the hydroxyl group (OH) on EG.[12] Then TBAA, a reactive azidation source[4] generated under these reaction conditions, promoted the reaction to obtain a GAP. The postulated mechanism for the synthesis of GAP under the system was depicted in Scheme 1. Through these results, it was observed that the DES of the TBAB-EG combination not only dissolves all the reactants well, but also have a dual effect of accelerating the reaction by generating a reactive azide source.

The reaction in the TBAB/EG system was investigated in detail, as shown in Table 3. PECH was completely converted to GAP in 8 h in both TBAB/EG and TBAC/EG, but TBAB was selected as the HBA component considering the ease of the workup process and economic feasibility (Table 3, entry 1). As expected, in TBAB or EG alone, the reaction conversion dropped to 29% and 9%, respectively, and the efficiency was much lower than that of the TBAB/EG condition (See SI). When the reaction temperature was lowered to 70 °C, the conversion was only 65% even after 24 h (Table 3, entry 2). As the third component of a DES, water has been reported to significantly affect the physicochemical properties of DESs, such as density, viscosity, and polarity. Therefore, we tested the impact of water content on TBAB/EG,[13] but it was ineffective (Table 3, entry 3). As the ratio of EG in the DES increased, the conversion rate was slower than when the molar ratio of TBAB/EG was 1:2; however, in both cases, the reaction was completely converted to GAP with stirring for 24 h (Table 3, entries 4 and 5). By increasing the concentration of TBAB/EG to 5 M or 10 M, PECH was completely converted to GAP within 5 h (Table 3, entries 6 and 7).

After reaction completion, water was poured into the reaction mixture and stirred under reflux. This washing process was repeated several times to purify GAP. Pure GAP was successfully isolated without using any organic solvents during the entire synthetic process and post-treatment (Figure 3). Moreover, during this workup procedure, the solvent could be

Table 3. Results of optimization of the azidation of PECh in TBAB/EG.[a]

Entry	DES (mole ratio)	Conc. (M)[c]	Temp.	Time	Conversion
1	TBAB/EG (1/2)	1.6	100 °C	8 h	>99%
2	TBAB/EG (1/2)	1.6	70 °C	24 h	65%
3[b]	[TBAB/EG]:H_2O	1.6	100 °C	5 h	84%
4	TBAB/EG (1/3)	1.6	100 °C	5 h	87%
				24 h	>99%
5	TBAB/EG (1/5)	1.6	100 °C	5 h	77%
				24 h	>99%
6	TBAB/EG (1/2)	5.0	100 °C	5 h	>99%
7	TBAB/EG (1/2)	10.0	100 °C	3 h	98%
				5 h	>99%

[a] All reactions were performed using PECH 300 mg (0.32 mmol), NaN_3 (1.2 equiv.) in TBAB/EG. [b] [TBAB/EG]:H_2O = 8/2 (v/v). [c] based on PECH.

© 2023 Wiley-VCH GmbH

23656549, 2023, 30, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/slct.202300703 by Zhejiang University, Wiley Online Library on [03/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

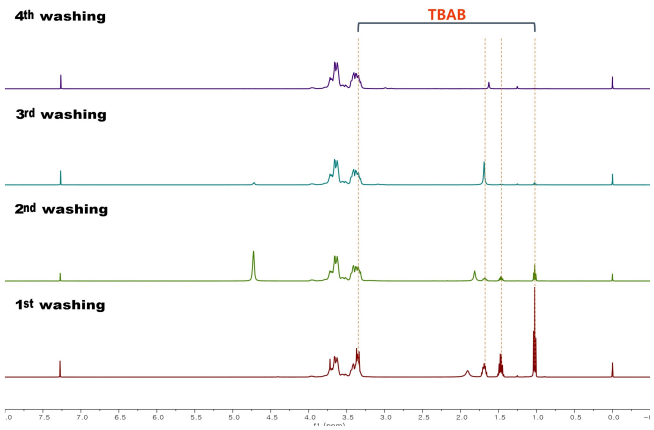
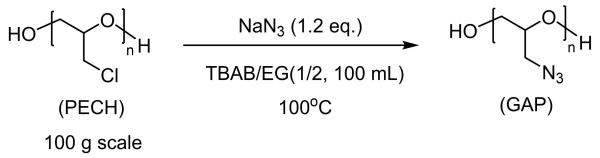


Figure 3. ^{1}H-NMR spectra of the GAP product after washing with water.

easily separated from the GAP and recovered by freeze-drying for the next run.

Encouraged by the above results, we increased the scale of the reaction to 100 g (1.08 M) of PECH in 100 mL of the TBAB/EG solvent system in a 500 mL-jacketed reactor, as outlined in Scheme 2 and Figure 4. Immediately after the completion of the reaction, the salts were successfully removed by hot filtration, and the obtained filtrates were washed with water several times to obtain pure GAP in the organic layer. The separated aqueous layer, including the solvent, was subjected to freeze drying to remove water and was recovered and reused in the next run. The recovered solvent was used to produce GAP in the 2nd run. Fortunately, the reaction proceeded smoothly, and the GAP product was obtained in good-to-excellent yields (1st: 93%, 2nd: 90%), indicating the possibility of solvent reusability.



Scheme 2. Scale-up for GAP synthesis.

Conclusions

In summary, for the first time, we have developed a facile and green synthetic method for GAP in TBAB-based DESs. This strategy avoids the use of organic solvents throughout the synthetic process, circumventing the disadvantages of organic solvents, such as toxicity, solvent waste, and health hazards. Moreover, it is very eco-friendly, as the DES can be recovered and reused. A simple workup and environmentally benign, safe process make this method an attractive and practical alternative to conventional GAP synthesis.

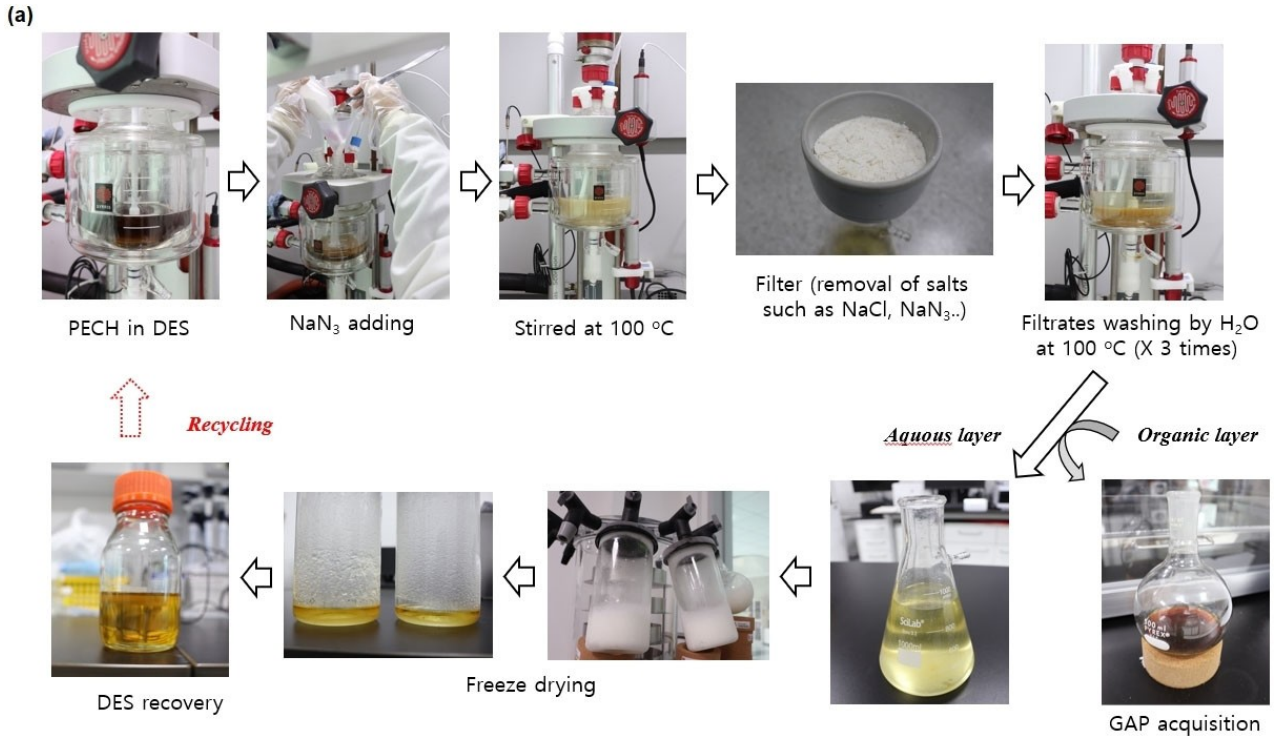


Figure 4. Scale-up procedure for GAP synthesis.

© 2023 Wiley-VCH GmbH

23656549, 2023, 30, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/slct.202300703 by Zhejiang University, Wiley Online Library on [03/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Experimental Section

General procedure for the preparation of DESs

The DESs were prepared according to procedures in the literature.[14] TBAB (396 g, 1.2 mol) and EG (150 g, 2.4 mol) were mixed in a round-bottomed flask at a molar ratio of 1:2, and the mixture was heated at 120 °C with stirring until a colorless homogeneous liquid formed. TBAB/EG (1/2) was allowed to cool to room temperature and was used directly for the synthesis of GAP without any further purification.

General procedure for the synthesis of GAP

To a solution of PECH (0.30 g, 3.2 mmol) in DES (2 mL) was added NaN_3 (0.26 g, 1.2 equiv.) at room temperature. The reaction mixture was stirred at 100 °C and monitored using FT-IR. After the completion of the reaction, water (4 mL) was added to the reaction mixture. The resulting mixture was stirred for 30 min at 100 °C. After separation of the layers, the organic layer was washed twice with water (4 mL) at 100 °C; this was repeated two times. The obtained organic layer was concentrated to give the pure GAP adduct (0.3 g, 93 %, amber syrup).

Scale-up procedure for the synthesis of GAP

To a solution of PECH (100 g, 1.08 mol) in TBAB/EG (molar ratio of 1:2, 100 mL) was added NaN_3 (84.8 g, 1.2 equiv.) at room temperature. The reaction mixture was then stirred at 100 °C for 5 h. After the completion of the reaction, the resultant mixture was filtered in a hot state. Water (300 mL) was added to the filtrate and stirred for 30 min at 100 °C. After separation of the layers, the organic fraction was washed twice with water (250 mL) at 100 °C; this was repeated two times. The aqueous fraction was evaporated by freeze drying, and TBAB/EG solvent was recovered and reused to give GAP in the 2nd run. The obtained organic layer was concentrated to give the pure GAP (99.9 g, 93 %, amber syrup).

Acknowledgements

We are grateful for support provided by the Agency for Defense Development.

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Keywords: Deep Eutectic Solvent · Azidation · PECH · GAP · Green Chemistry

[1] a) M. B. Frankel, L. R. Grant, J. E. Flanagan, *J. Propul. Power* **1992**, *8*, 560–563; b) K. V. Puduppakkam, M. W. Beckstead, *Combust. Sci. Technol.* **2005**, *177*, 1661–169.
[2] a) B. Gaur, B. Lochab, V. Choudhary, I. K. Varma, *J. Macromol. Sci. Polym. Rev.* **2003**, *C43*, 505–545; b) Y. M. Mohan, M. P. Raju, K. M. Raju, *J. Appl. Polym. Sci.* **2004**, *93*, 2157–2163; c) J.-G. Wang, D. Liu, G.-Y. Cao, J.-X. Li, *Chem. Propellants Polym. Mater.* **2007**, *5*, 16–23; d) M. Abd-Elghany, T. M. Klapötke, A. Elbeih, S. Zeman, *J. Anal. Appl. Pyrolysis* **2017**, *126*, 267–274; e) T. Cheng, *Des. Monomers Polym.* **2019**, *22*, 54–65.
[3] a) E. J. Vandenberg, W. Foulk, US 3645917, **1972**; b) G. Ampleman, US 5124463, **1992**; c) G. Ampleman, US 5256804, **1993**; d) G. Ampleman, US 5359012, **1994**; e) S. P. Ribeiro, D. G. Santiago, A. S. Vianna Jr, *Polimeros* **2012**, *22*, 407–413; f) S. P. Ribeiro, D. G. Santiago, A. S. Vianna, Jr., *Polimeros* **2012**, *22*, 414–421; g) C. Mura, S. Fruci, P. Lamia, M. Cappello, S. Filippi, G. Polacco, *J. Energ. Mater.* **2016**, *34*, 216–233.
[4] R. I. Wagner, US 5055600, **1991**.
[5] X. Xu, M. Liu, Y. Yin, C. Zheng, P. Denga, D. Xuea, *Green Chem.* **2016**, *18*, 1364–1367.
[6] A. Kasztankiewicz, A. Kogut, M. Jankiewicz, P. Maksimowski, *Problems of Mechatronics. Armament, Aviation, Safety Engineering* **2018**, *9*, 59–72.
[7] a) A. P. Abbott, G. Capper, D. L. Davies, R. K. Rasheed, V. Tambyrajah, *Chem. Commun.* **2003**, 70–71; b) E. L. Smith, A. P. Abbott, K. S. Ryder, *Chem. Rev.* **2014**, *114*, 11060–11082; c) L. J. B. M. Kollau, M. Vis, A. van den Bruinhorst, A. C. C. Esteves, R. Tuinier, *Chem. Commun.* **2018**, *54*, 13351–13354.
[8] a) C. Ruß, B. König, *Green Chem.* **2012**, *14*, 2969–2982; b) S. E. Hooshmand, R. Afshari, D. J. Ramón, R. S. Varma, *Green Chem.* **2020**, *22*, 3668–3692; c) B. B. Hansen, S. Spittle, B. Chen, D. Poe, Y. Zhang, J. M. Klein, A. Horton, L. Adhikari, T. Zelovich, B. W. Doherty, B. Gurkan, E. J. Maginn, A. Ragauskas, M. Dadmun, T. A. Zawodzinski, G. A. Baker, M. E. Tuckerman, R. F. Savinell, J. R. Sangoro, *Chem. Rev.* **2021**, *121*, 1232–1285.
[9] a) D. Carriazo, M. C. Serrano, M. C. Gutiérrez, M. L. Ferrer, F. del Monte, *Chem. Soc. Rev.* **2012**, *41*, 4996–5014; b) F. del Monte, D. Carriazo, M. C. Serrano, M. C. Gutiérrez, M. L. Ferrer, *ChemSusChem* **2014**, *7*, 999–1009; c) F. Del Monte, D. Carriazo, M. C. Serrano, M. C. Gutiérrez, M. L. Ferrer, *Chem. Sus. Chem.* **2014**, *7*, 4, 999–1009; d) J. D. Mota-Morales, R. J. Sánchez-Leija, A. Carranza, J. A. Pojman, F. del Monte, *Prog. Polym. Sci.* **2018**, *78*, 139–153.
[10] J.-A. Shin, S. Lee, *ChemistrySelect* **2022**, *7*, e202202000.
[11] a) A. Pandey, R. Rai, M. Pal, S. Pandey, *Phys. Chem. Chem. Phys.* **2014**, *16*, 1559–1568; b) A. Pandey, S. Pandey, *J. Phys. Chem. B* **2014**, *118*, 14652–14661; c) C. Florindo, A. J. S. McIntosh, T. Welton, L. C. Branco, I. M. Marrucho, *Phys. Chem. Chem. Phys.* **2018**, *20*, 206–213.
[12] R. Yusof, K. Jumbri, M. B. A. Rahman, *J. Mol. Liq.* **2021**, *339*, 116709–116720.
[13] a) C. Florindo, F. S. Oliveira, L. P. N. Rebelo, A. M. Fernandes, I. M. Marrucho, *ACS Sustainable Chem. Eng.* **2014**, *2*, 2416–2425; b) R. Craveiro, I. Aroso, V. Flammia, T. Carvalho, M. T. Viciosa, M. Dionio, *J. Mol. Liq.* **2016**, *215*, 534–540; c) D. O. Abranches, L. P. Silva, M. A. R. Martins, J. A. P. Coutinho, *J. Chem. Phys.* **2021**, *155*, 34501–34508.
[14] A. P. Abbott, R. C. Harris, K. S. Ryder, C. D. Agostino, L. F. Gladden, M. D. Mantle, *Green Chem.* **2011**, *13*, 82–90.

Manuscript received: February 23, 2023

© 2023 Wiley-VCH GmbH

23656549, 2023, 30, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/slct.202300703 by Zhejiang University, Wiley Online Library on [03/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License